

Docu-Vision

For



2 0 2 4



Hello,

We are **MS_Cube : **M**eghana, **S**anjana, **S**hih-Hsuan, **S**anthosh**

A team of Master's students in Business Analytics dedicated to solving real-world challenges through innovative technology.

Overview



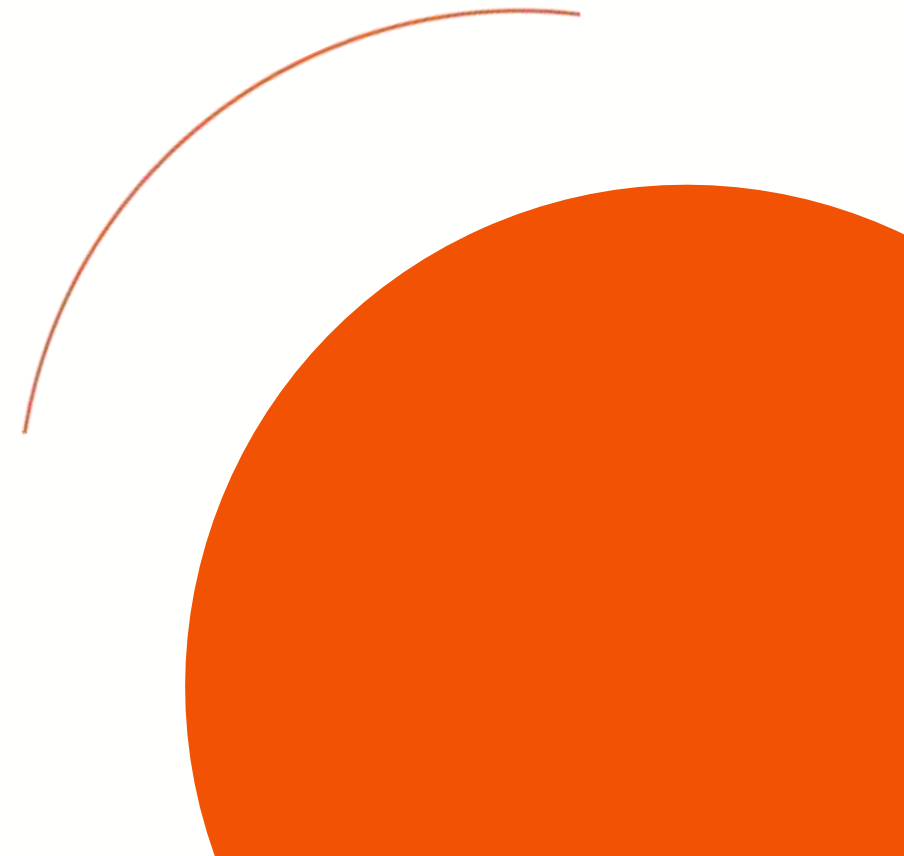
1. Problem Statement

2. Act 1: The Rise

3. Act 2: The Fall

4. Act 3: The Resurrection

5. Results



Problem Statement:



- Unstructured document processing relies heavily on OCR accuracy, limited by its error-handling capabilities and longer processing times, hampering efficiency in financial operations.
- This project aims to leverage multimodal Large Language Models to create a robust information extraction pipeline, bypassing OCR limitations and enhancing accuracy across various document formats.

Models: GPT-4o

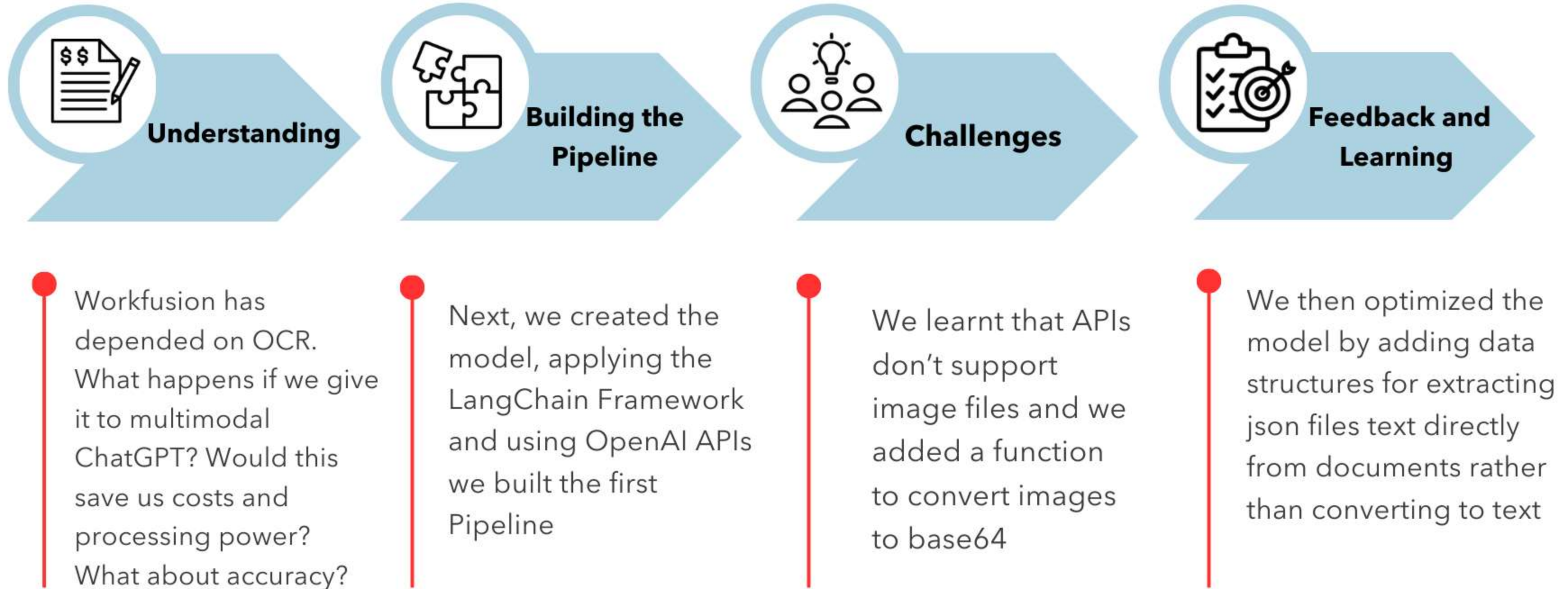
Datasets:

- 1. Invoices** – a Total of 50 documents with formats of .pdf, .jpg, .tiff, .jpeg
- 2. Articles of Incorporation** – a total of 9 .pdf documents with a total of 21 pages

Act I : The Rise



Research and Learnings



Prompt

Complete Prompt:

```
[{"role": "system", "content": f"{system_prompt}"},
 {"role": "user", "content": [{"type": "text", "text": f"{user_prompt}"},
                               {"type": "image_url",
                                "image_url": {"url": f"data:image/jpeg;base64,{base64_image}"}}
                               ]
}]
```

User Prompt:

```
There are two types of entities to extract from the given images table and non-table.
non-table labels are receiver_name,invoice_date,invoice_number,tax_amount1,grand_total1.
table labels are quantity_per_item1, price_per_item1, total_per_item1.
The height = 841 and width=595 of the image is consider the image dimension while returnng bbox of
extracted entities.
output format:
{'non-table': [{'entity': name of non-table entity, 'text': extracted value for name entity if text is
multiline then eachline should be seperated by
else just return the extracted text,'bbox':[left, top, right, bottom]}], 'table': [{'entity':name of table
entity, 'text': extracted value for name entity if text is multiline then eachline should be seperated
by
else just return the extracted text, 'bbox': [left, top, right, bottom], 'row-number': row number of the
table}]}
Important Instruction:
> Always try your best and Only extract the text for a given label and return in output if you are very
sure about it else do not return that label in output response.
> If a label has multiple values the you MUST return each value as individual extracted text.
> bounding boxes of the extracted entities are very important. YOU SHOULD RETURN THE EXACT bbox of
extracted entities.
> Always return valid bounding boxes, MUST NOT return [0, 0, 0, 0] or empty text for extracted entity.
```

System Prompt:

You are an advanced document information extraction model specializing in extracting structured information from invoices. Your goal is to extract specific given fields from an invoice image with its bounding boxes. Bounding boxes must be normalized from 0 to 1.

JSON Structure – List of Dictionaries

Articles of Incorporation

```
json
{
  "non-table": [
    {
      "entity": "corporation_name",
      "text": "Fahey Inc",
      "bbox": [0.244, 0.091, 0.344, 0.105]
    },
    {
      "entity": "corporation_agent",
      "text": "Mrs. Olen O'Hara",
      "bbox": [0.076, 0.409, 0.196, 0.423]
    },
    {
      "entity": "corporation_agent_address",
      "text": "24262 Greenholt Ridges\nSouth Caleb\nCALIFORNIA 900##",
      "bbox": [0.076, 0.432, 0.344, 0.473]
    },
    {
      "entity": "total_number_of_shares",
      "text": "94",
      "bbox": [0.576, 0.545, 0.596, 0.559]
    },
    {
      "entity": "incorporator_name",
      "text": "Ivonne McCullough",
      "bbox": [0.076, 0.682, 0.216, 0.696]
    }
  ],
  "table": []
}
```

Invoice

```
json
{
  "non-table": [
    {
      "entity": "invoice_number",
      "text": "123456",
      "bbox": [0.579, 0.173, 0.683, 0.213]
    },
    {
      "entity": "invoice_date",
      "text": "5/1/2014",
      "bbox": [0.751, 0.173, 0.855, 0.213]
    },
    {
      "entity": "grand_total1",
      "text": "551.56",
      "bbox": [0.845, 0.853, 0.947, 0.893]
    }
  ],
  "table": [
    {
      "entity": "total_per_item1",
      "text": "200.00",
      "bbox": [0.845, 0.493, 0.947, 0.533],
      "row-number": 1
    },
    {
      "entity": "total_per_item1",
      "text": "375.00",
      "bbox": [0.845, 0.547, 0.947, 0.587],
      "row-number": 2
    },
    {
      "entity": "total_per_item1",
      "text": "(50.00)",
      "bbox": [0.845, 0.600, 0.947, 0.640],
      "row-number": 3
    },
    {
      "entity": "total_per_item1",
      "text": "28.56",
      "bbox": [0.845, 0.653, 0.947, 0.693],
      "row-number": 4
    }
  ]
}
```

Act II : The Fall



THE BOTTLE NECK.....INCONSISTENT RESULTS!

INVOICE

February 10, 2022

#40

Due: March 12, 2022
Project: Pest control
Description: Consulting and supplies for the eradication of unwanted birds

Wile E. Coyote
1 Monument Valley
Arizona, United States
wile.e.coyote@latextemplates.com

DESCRIPTION	QUANTITY	UNIT PRICE	SUBTOTAL	
Heavy wrought anvil	1.00	\$5,600.00	\$5,600.00	
Dynamite (6pk)	25.00	\$24.99	\$624.75	
ACME TNT detonator	1.00	\$0.00	\$0.00	Complimentary
Hi-speed tonic	10.00	\$102.99	\$1,029.90	
Do-it yourself tornado kit	1.00	\$3,495.00	\$3,495.00	
Refund for defective rocket	1.00	-\$1,000.00	-\$1,000.00	Final settlement
ACME Consulting: project planning and pest control ornithologist consulting	23.50	\$120.00	\$2,820.00	
			BEFORE TAX	\$12,569.65
			TAX (12.5%)	\$1,571.21
			TOTAL (USD)	\$14,140.86

Terms and Conditions: Products sold by ACME Corporation come with no guarantees or warranties of any kind, expressed or implied. ACME specifically disclaims all implied warranties of any kind or nature, including any implied warranty of merchantability and/or any implied warranty of fitness for a particular purpose.

ACME Corporation
2295 Horseneck Road
Fairfield, NJ 07004
United States

Contact
ACME.com
sales@acme.com

Payment
Bank of the United States
Sort Code: 010-110
Account: 10011001110

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x_min , y_min , x_max, y_max

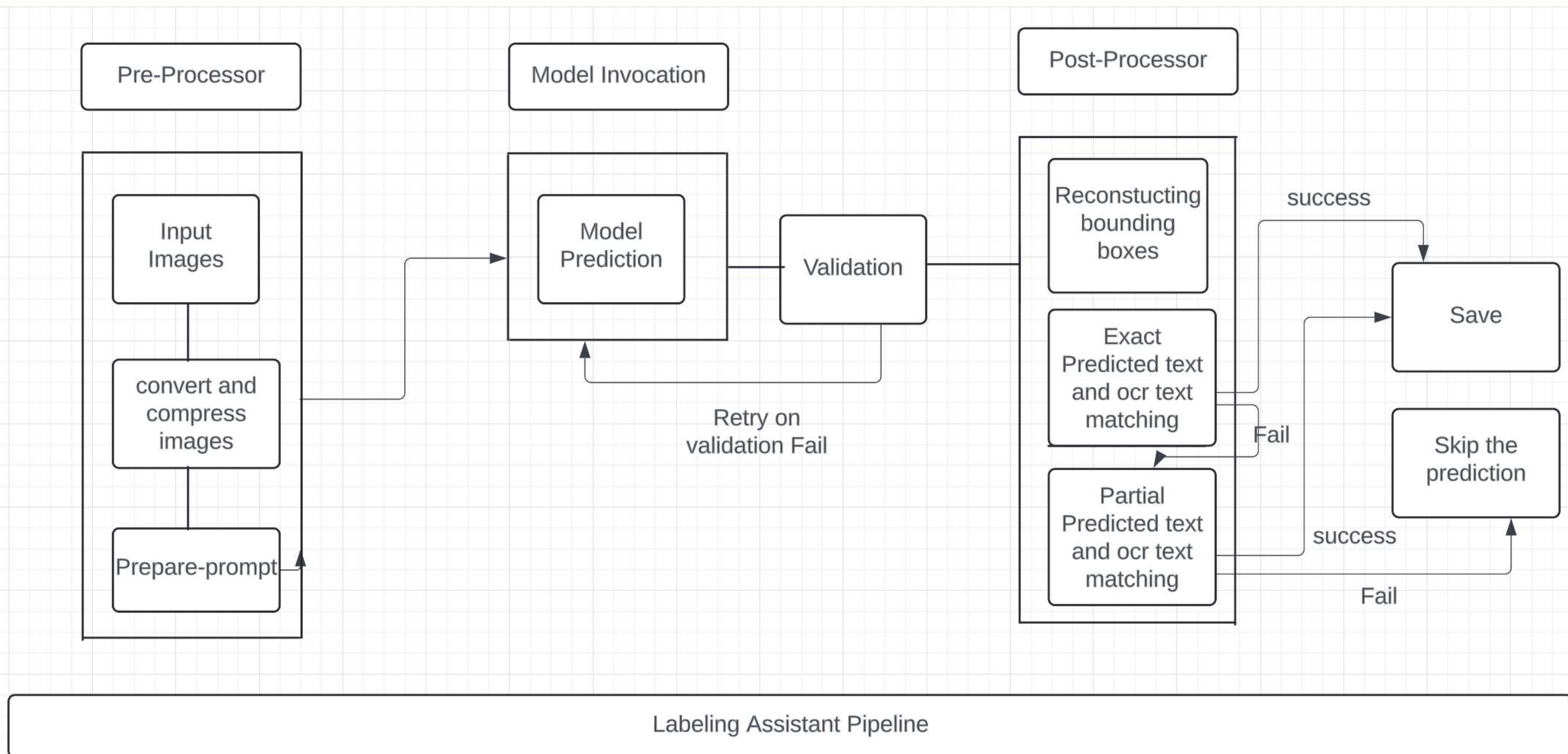
Left, top, bottom, right

normalised left, top, bottom, right

Act III : The Resurrection



Workflow



Results and Statistics

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Invoice Per-field Statistics:

FIELD	PRECISION	RECALL	F1	# of samples
invoice_date	96%	94%	95%	49
price_per_item	91%	90%	91%	140
quantity_per_item	91%	88%	90%	145
invoice_number	82%	89%	85%	44
total_per_item	78%	82%	80%	130
grand_total	69%	67%	68%	48
tax_amount	60%	54%	57%	28
invoiced_to	45%	43%	44%	47
AVERAGE	77%	76%	76%	

Invoice

Relationship Statistics:

Entity	PRECISION	RECALL	F1	Support
Invoice	81%	79%	80%	370

CONCLUSION:

- All the green highlighted fields are the fields that have higher accuracies while red highlighted fields need improvement.
- Fields such as **tax_amount** and **invoiced_to** need to be improved.
- The model demonstrates strong performance with an **F1 score of 0.80**, balancing **high precision (0.81)** and **solid recall (0.79)** in extracting structured information from invoices across 370 ground truth relationships. While reliable, the slight gap between precision and recall suggests opportunities to improve detection of subtle or missed relationships.

Invoice Per-field Statistics Comparison with BaseLine:

FIELD	Baseline F1(A)	Current F1(B)	B-A	# of samples
grand_total	43%	68%	+25%	48
price_per_item	81%	91%	+10%	140
invoice_number	77%	85%	+8%	44
tax_amount	49%	57%	+8%	28
quantity_per_item	85%	90%	+5%	145
invoice_date	92%	95%	+3%	49
total_per_item	77%	80%	+3%	130
invoiced_to	50%	44%	-6%	47
Average	69%	76%	+7%	

CONCLUSION:

- On average there is a 7% improvement in per-field statistics for Invoice dataset.
- However, field like invoiced_to has a significant drop while fields such as grand_total has significant improvement.
- These can be because of failure of reconstruction of bounding boxes due to OCR errors.

Invoice Relational Statistics Comparison with BaseLine:

Doc Type	Baseline F1(A)	Current F1(B)	B-A	Support
Invoice	77%	80%	+3%	370

- On average, there is a 3% improvement in relational statistics for Invoice dataset.

Articles of Incorporation (AOI) Per-field Statistics:

FIELDS	PRECISION	RECALL	F1	# of samples
total_number_of_shares	100%	100%	100%	5
incorporator_name	81%	83%	82%	10
corporation_name	70%	78%	74%	9
corporation_agent	66%	70%	68%	10
date_of_signature	65%	60%	62%	5
shareholder_names	58%	54%	56%	13
period_of_corporation_duration	43%	67%	52%	3
corporation_address	44%	53%	48%	5
corporation_agent_address	21%	33%	26%	6
incorporator_address1	17%	25%	20%	4
AVERAGE	59%	59%	59%	

CONCLUSION:

- All the fields in the green are doing considerably well, while red highlighted fields need to be improved.
- Long fields like addresses are mostly having low scores.

AOI Statistics Comparison with BaseLine:

Fields	Baseline F1 (A)	Current F1 (B)	B - A	# of samples
shareholder_names	27%	56%	+29%	13
date_of_signature	39%	62%	+23%	5
total_number_of_shares	83%	100%	+17%	5
period_of_corporation_duration	40%	52%	+12%	3
corporation_agent	63%	68%	+5%	10
incorporator_address1	22%	20%	-2%	4
incorporator_name	84%	82%	-2%	10
corporation_name	100%	74%	-26%	9
corporation_agent_address	55%	26%	-29%	6
corporation_address	80%	48%	-32%	5
AVERAGE	59%	59%	+0%	

CONCLUSION:

- On average there is no statistics change in AOI dataset.
- However, long texts like addresses mostly have significant drops with new approach while some fields like shareholder_names, date_of_signature and total_number_of_shares has significant improvements.
- These can be because of failure of reconstruction of bounding boxes due to OCR errors.
- RAG applied for baseline to avoid splitting pages which can be the reason to get incorrect prediction because of losing information. We can try to send two page at a same time to get better results in current

Error Analysis

California Articles of Incorporation (Close Corporation)
ARTICLES OF INCORPORATION

I.
The name of this corporation is Pagac Inc

II.
The purpose of the corporation is to engage in any lawful act or activity for which a corporation may be organized under the GENERAL CORPORATION LAW of California other than the banking business, the trust company business or the practice of a profession permitted to be incorporated by the California Corporations Code.

III.
The name and address in the State of California of this corporation's initial agent for service of process is:
Name Isaias Bartoletti
Address 5193 Reynaldo Prairie
STREET Address (DO NOT USE POST OFFICE BOX)
City Schaeferia State CALIFORNIA Zip 900##

IV.
This corporation is authorized to issue only one class of shares of stock; and the total number of shares which this corporation is authorized to issue is 95.

V.
This corporation is a CLOSE CORPORATION. All of the corporation's issued shares of stock, of all classes, shall be held of record by not more than 10 persons.

(Signature of Incorporator)
Joel Nolan,
Incorporator

INSTRUCTIONS:
To incorporate utilizing the attached form, it will be necessary for you to complete the form with the required information. The information must be typed with letters in dark contrast to the paper. Documents submitted which would produce poor microfilm will be returned unfilled. Articles of Incorporation may be drafted to include all required provisions and any statutory permissive provisions, including initial directors. The Secretary of State does not have samples containing permissive provisions.
Article I—is to be completed with the name of the corporation exactly as the name is to appear on the records of the Secretary of State. The name of a close corporation must include the word "corporation", "incorporated" or "limited" or an abbreviation of one of the three.
Article II—has been taken directly from the California Corporations Code as is required by law and should not be modified when drafting documents.
Article III—is to be completed with the name and California street address, or the physical location, of the agent for service of process a post office box is not acceptable). The designated agent, individual or corporation, must agree to accept process on behalf of the corporation prior to designation. A proposed corporation cannot designate itself as agent for service of process. When designating another corporation as agent, the agent corporation must have on file, in the Secretary of State, a statement pursuant to Section 1505, California Corporation Code. When a corporate agent is used, the address of the designated corporation is to be omitted from the articles.
Article IV—is to be completed with the total number of shares that the corporation is authorized to issue.
Article V—is to be retained as stated with the number of persons completed. Please note that the number of persons designated cannot exceed 35.
The Articles of Incorporation must be originally signed by an incorporator unless initial directors are named in the articles. If initial directors have been named in the Articles of Incorporation the directors named must sign and acknowledge the document in lieu of an incorporator. The name(s) of the person(s) signing must be typed directly below the signature.

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- GPT often **corrects or reinterprets content**, while the golds has the output text as is, without adjustments. (For corporation_agent_address)

- Both OCR and GPT occasionally fail to capture text accurately, especially when parts of the **text are cut or incomplete**. (For corporation_name)

Error Analysis

and the corporation's mailing address is:

163 Sipes Mission

West Traceyberg, NJ, 09285-2139

Article VII Authorized Shares

The number of shares of stock the corporation is authorized to issue is: 51.

The class of stock issued shall be Class F stock.

Each share shall have a par value of \$ 92.

Article VIII Directors & Officers

The initial directors and officers shall be the following persons and positions:

Director Shantel Sporer	Address Suite 454 7600 Bogisich Mountain, Ardenburgh, CT 218
Director	Address
Director	Address
President Reanna Parker	Address Apt. 963 48698 Hintz Hollow, Kiethhaven, CO 09075
Secretary Teri Pfannerstill	Address Apt. 633 1199 Mraz Bypass, Briarside, PA 53102
Treasurer Gilda Wolff	Address 6773 Vandervort Hollow, South Gertrudestad, TN 65765

Article IX Bylaws

The Incorporator shall adopt the initial bylaws of the corporation. The shareholders may amend the bylaws at anytime by the provisions provided therein.

Article X Dissolution

Upon dissolution, assets shall be distributed by the board of directors according to the applicable statutes and regulations for the State of Minnesota. Further provisions regarding distribution upon dissolution shall be stated in the corporation's bylaws.

Article XI Indemnification

The corporation does indemnify any directors, officers, employees, incorporators, and shareholders of the corporation from any liability regarding the corporation and the business of the corporation, unless

Corporate Filing and Services Division

1500 Eleventh Street
Sacramento, CA 95814
Attention: Document Filing Support Unit
(916) 657-5448

OR

may be hand delivered for over the counter processing to any of the corporate branch offices which are located in:

—Fresno (209) 243-2100
2497 West Shaw, Suite 101
Fresno, CA 93711
—Los Angeles (213) 897-3062
107 South Broadway, Suite 4001
Los Angeles, CA 90012-4593
—San Diego (619) 525-4113
1350 Front Street, Suite 2060
San Diego, CA 92101-3690

- **Incorrect prediction** as we are passing page by page which can be a reason to lose **some contextual information**.

- The gold label (sometimes) does not contain corporation_address sometimes and GPT tends to **interpret it from the document**. OCR may need tags when addresses are unclear.

Error Analysis

SALES INVOICE TEMPLATE

Company Name

123 Main Street

Hamilton, OH 44416

(321) 456-7890

Email Address

DATE

02/15/18

INVOICE NO.

A246

CUSTOMER NO.

114H

BILL TO

ATTN: Name / Dept

Company Name

123 Main Street

Hamilton, OH 44416

(321) 456-7890

Email Address

SHIP TO

ATTN: Name / Dept

Company Name

123 Main Street

Hamilton, OH 44416

(321) 456-7890

P.O. NO.	SHIP DATE	SHIP VIA	SALESPERSON	FOB	TERMS

ITEM NO.	DESCRIPTION	QTY	UNIT PRICE	TOTAL
A111	Women's Tall - M	10	\$ 10.00	\$ 100.00
B222	Men's Tall - M	5	\$ 20.00	\$ 100.00
C333	Children's - S	10	\$ 5.00	\$ 50.00
D444	Men's - XL	5	\$ 10.00	\$ 50.00
			\$ -	\$ -
			\$ -	\$ -
			\$ -	\$ -
			\$ -	\$ -

Remarks / Instructions:

SUBTOTAL

\$ 300.00

TAX RATE

3.80%

\$ 11.40

SHIPPING/HANDLING

\$ 30.00

OTHER

\$ 15.00

TOTAL

\$ 356.40

Please make check payable to Your Company Name.

THANK YOU

For questions concerning this invoice, please contact

Name, (321) 456-7890, Email Address

www.yourwebaddress.com

Some Considerations Based on Observation:

- OCR and GPT limitations:** Both have issues with text extraction accuracy.
- Clear address formatting:** Improves **OCR** and **GPT** performance.
- Use of tags:** Helps **OCR** capture data when addresses are unclear.
- Proximity of symbols:** Ensure \$ **symbol** is close to its field for **GPT** accuracy.
- Regular validation:** Ensures better data accuracy from both **OCR** and **GPT**.

- GPT may capture the address from the invoiced_to field ,if it's placed below the name, and sometimes fails to capture the \$ symbol if it's too far from the field.

Cost Analysis

	Baseline			Current	
Datasets	Input Cost	Output Cost	Inference Time	Output Cost	Inference Time
Invoices (50 Pages)	\$ 0.22	\$ 0.22	9 sec /call	\$ 0.35	10 sec /call
AOI (21 Pages)	\$ 0.3	\$ 0.03	8 sec /call	\$ 0.04	6 sec / call

Unit Price : \$2.50 / 1 M input Tokens
\$10.00 / 1 M Output Tokens

****Note:** OpenAI multimodal calculates input cost based on input tokens, image size, and resolution. The image tokenizer most likely has not been published yet; the unit price has been provided for price estimation.

Conclusion: The cost of RAG has not been added for baseline cost analysis. The output cost difference is not significant between both approaches but we may save some cost for input since we are sending images instead of OCR texts to model but need some further research on OpenAI image to token calculation.

Conclusion:

- Information and relation extraction with a multimodal model can be a potential option to eliminate OCR cost and OCR-related errors. It does have similar statistics.
- Unfortunately, at this moment, OCR has been used since the model does not predict accurate bounding boxes, and OCR text matching has been used to reconstruct the bounding boxes, which is essential for information and relation extraction.
- Potentially minimizing cost by sending image instead of huge number of OCR tokens and RAG to retrieve relevant context for model input in previous labeling assistant pipeline.

Next Steps:

- Improve prompt or extraction logic to get a better model score.
- Try different vendors with multimodal models like (OpenAI o1) to improve the results.
- Test different multimodal models to find which ones give us the correct bounding boxes.
- Few-shot Learning can be an option (needed some research)
- Leverage Open-Source Models for better control over contextual understanding of the processing document

Thank you

